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701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Gut Microbiome Diversity and Composition Changes Post Allogeneic Hematopoietic Stem Cell Transplant (AlloSCT): A Single-Center Prospective Analysis

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AIM The aim of this study is to evaluate the impact of AlloSCT on gut microbial diversity and its composition at our institution.

Background Hematopoietic stem cell transplantation (HSCT) often leads to a loss of gut microbiome diversity due to factors such as the conditioning regimen, antibiotic use, and impaired dietary intake. Post-transplant, the gut microbiome can shift towards an increase in enteropathogens and a decrease in beneficial commensal bacteria.

Study Design We conducted a single-center prospective study at the University of Arkansas for Medical Sciences (UAMS) in AlloSCT patients (pts) approved by our institution's IRB. Baseline stool samples were collected either at admission, within 24-72 hours prior to conditioning chemotherapy, or with in 48 hours post-chemotherapy. Follow-up stool samples were collected during the peri-engraftment period between days 12-24 to assess changes in gut microbiome diversity. Using V4-V5 primers for 16S rRNA sequencing, we analyzed differences in gut microbiota composition during the course of the transplant. Gut microbial phylogeny was constructed via IQ-tree, and diversity was measured using alpha diversity indices (Faith's phylogenetic diversity and Shannon index) and beta diversity indices (Bray-Curtis and Jaccard methods). Patient charts were reviewed to assess clinical outcomes, including overall survival, Graft-versus-host disease (GVHD), and relapse rates. All patients received standardized prophylactic antibiotics, including fluoroquinolones, azoles during the neutropenic phase, acyclovir for herpes simplex virus prophylaxis, and treatment for neutropenic fever per institution protocol during their transplant admission.

Results We analyzed 13 patients with a median age of 62 years (range, 44-74). Of these, 30.8% (4/13) were female and 69.2% (9/13) were male, with 76.9% (10/13) identified as Caucasian and 23.1% (3/13) as other ethnicities. Eight patients had acute myeloid leukemia (AML), three had myelodysplastic syndrome (MDS), and two had acute lymphoblastic leukemia (ALL). Of the patients included, 46.5% (6/13) had matched unrelated donors, 38.5% (5/13) had haploidentical donors, 7.7% (1/13) had matched related donors, and 7.7% (1/13) had mismatched unrelated donors. Myeloablative GVHD regimens were administered to 38.5% (5/13) of patients, while 61.5% (8/13) received reduced-intensity chemotherapy (RIC). GVHD prophylaxis included post-transplant cyclophosphamide (PTCy) + mycophenolate + tacrolimus for 84.6% (11/13) of patients, and Campath + tacrolimus for 15.4% (2/13).

Our study showed that engraftment induced a significant loss in gut microbial alpha diversity as measured by both Faith's phylogenetic diversity (baseline 30.8, engraftment 14.2; $p \leq 0.0007$) and Shannon's index (baseline 3.6? vs engraftment 2.4; $p < 0.05$). Gut microbial beta diversity also revealed significant differences between the baseline and engraftment groups: Bray-Curtis ($p \leq 0.03$), Jaccard index (baseline ($p \leq 0.004$). We also observed significant differences in enrichment of seven microbial features based on (CLR transformed) relative abundance. With loss of obligate anaerobes bacterial genus between baseline and engraftment stool samples - Clostridium innocuum group (1.5 vs 0.1), Blautia (5 vs 0), Anaerotruncus (1.8 vs 0), UBA1819 (3.1 VS 0) and gain of gram negative pathogenic bacterial genus enterococcus (-1 vs 0.5); $p < 0.05$.

Median progression-free survival and overall survival post-AlloSCT were 9 months and 12 months, respectively. The cumulative incidence of acute lower gastrointestinal graft-versus-host disease was 38.5% . Median follow-up in our study was 12.5 months.

Conclusion Our study highlights a significant decrease in gut microbial diversity and an alteration in gut microbial composition post-AlloSCT, with a loss of commensal obligate anaerobes and a gain of gram-negative enteropathogens. Due to the small

sample size, the impact on clinical outcomes such as graft-versus-host disease and transplant-related mortality post-AlloSCT remains uncertain. A prospective study is ongoing to assess the impact of gut microbial diversity on AlloSCT outcomes.

Disclosures No relevant conflicts of interest to declare.

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